# EXPERIMENT : 16

AIM:

Experiment to understand EDA-Quantitative and Qualitative analysis.

DESCRIPTION:

Understand the importance of EDA-Quantitative and Qualitative analysis.

CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

%matplotlib inline

# Load the tips dataset
tips = sns.load_dataset('tips')
print(tips.head())

# Distribution plots
sns.displot(data=tips, x='total_bill', kde=True)
plt.show()

sns.displot(data=tips, x='total_bill', kde=False)
plt.show()

# Joint plots
sns.jointplot(data=tips, x='tip', y='total_bill')
plt.show()

sns.jointplot(data=tips, x='tip', y='total_bill', kind='reg')
plt.show()

sns.jointplot(data=tips, x='tip', y='total_bill', kind='hex')
plt.show()
```

```
# Pair plots
sns.pairplot(tips)
plt.show()

print(tips['time'].value_counts())

sns.pairplot(tips, hue='time')
plt.show()

sns.pairplot(tips, hue='day')
plt.show()

# Heatmap
sns.heatmap(
    tips.corr(numeric_only=True),
    annot=True
)
plt.show()

# Box plots
sns.boxplot(x=tips['total_bill'])
plt.show()

sns.boxplot(x=tips['tip'])
plt.show()

# Count plots
sns.countplot(data=tips, x='day')
plt.show()

sns.countplot(data=tips, x='sex')
plt.show()

# Pie chart
tips['sex'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()

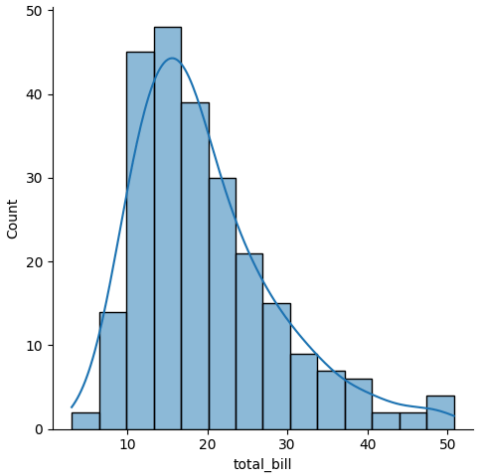
# Bar chart
tips['sex'].value_counts().plot(kind='bar')
plt.show()

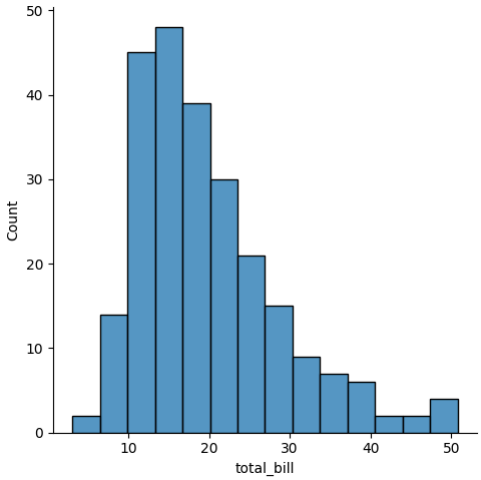
# Count plot for Dinner customers by day
```

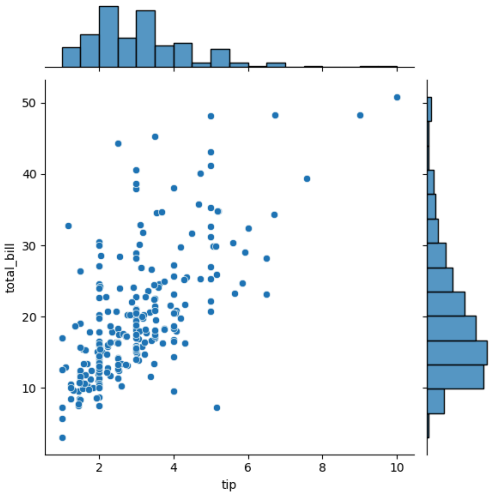
```
sns.countplot(
    data=tips[tips['time'] == 'Dinner'],
    x='day'
)
plt.show()
```

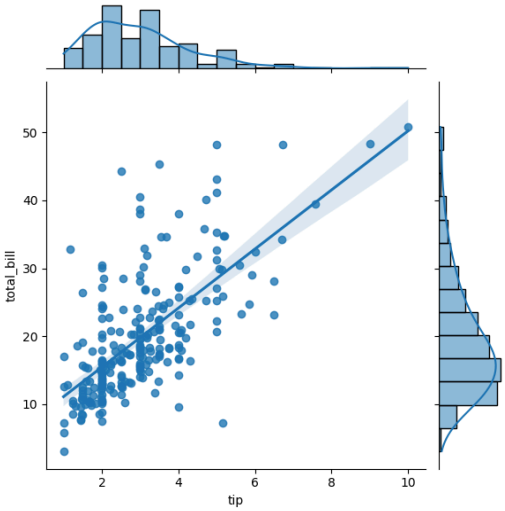
OUTPUT:

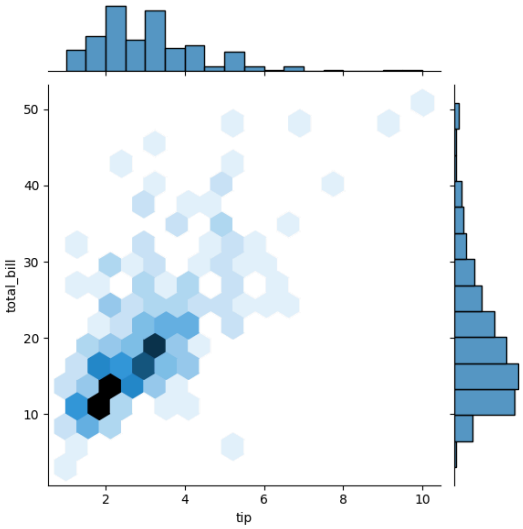
```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
```

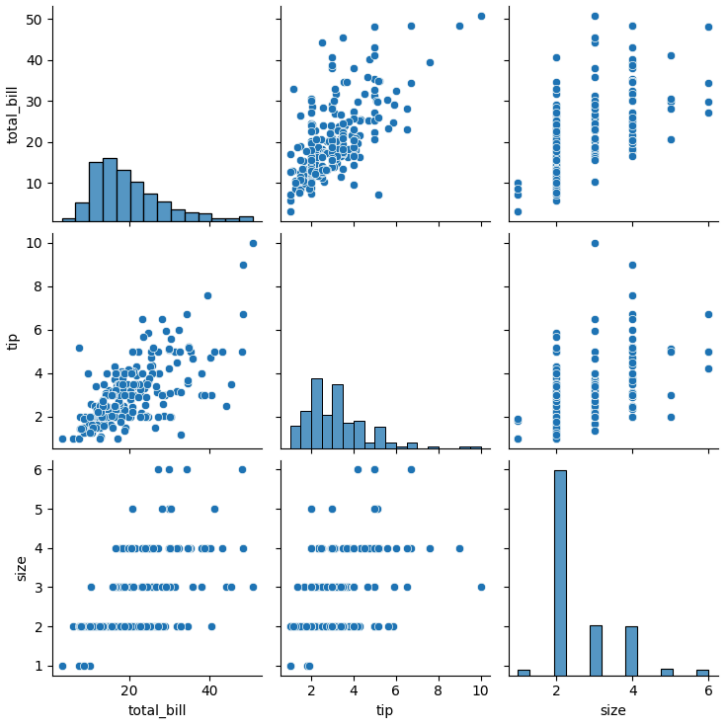


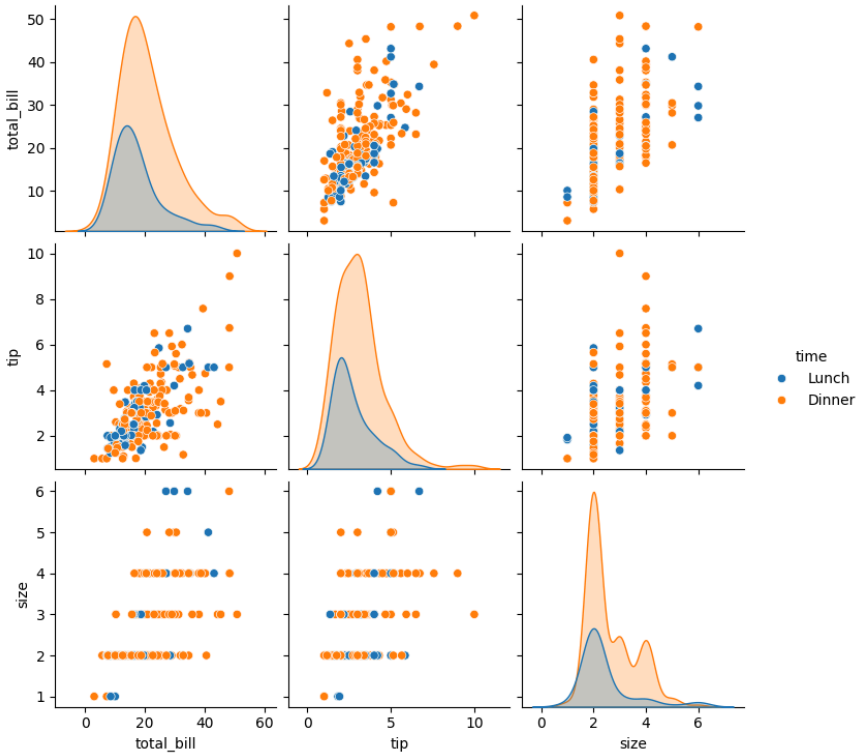

total_bill
tip
10
20
30
40
50
2
4
6
8
10

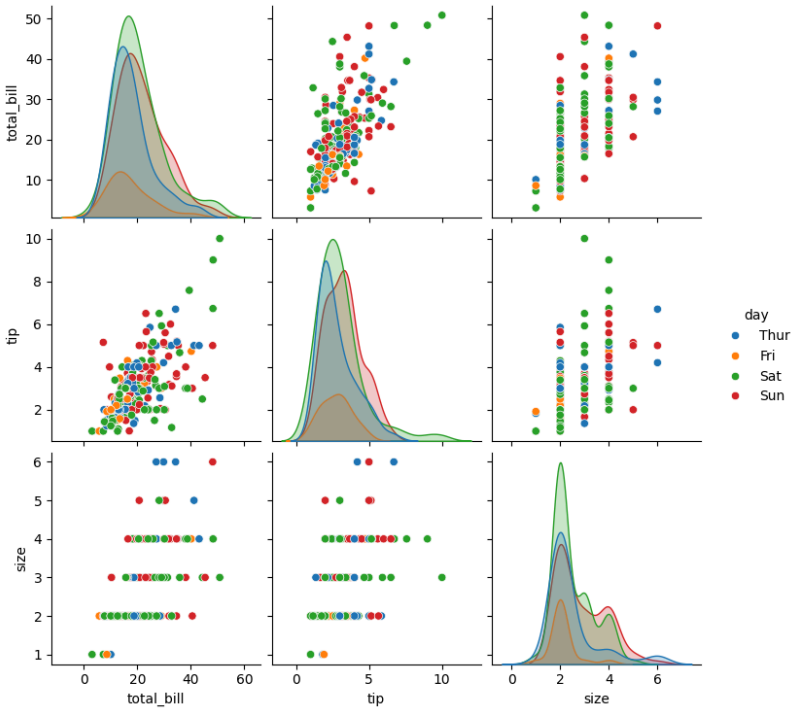

total_bill
tip
10
20
30
40
50
2
4
6
8
10

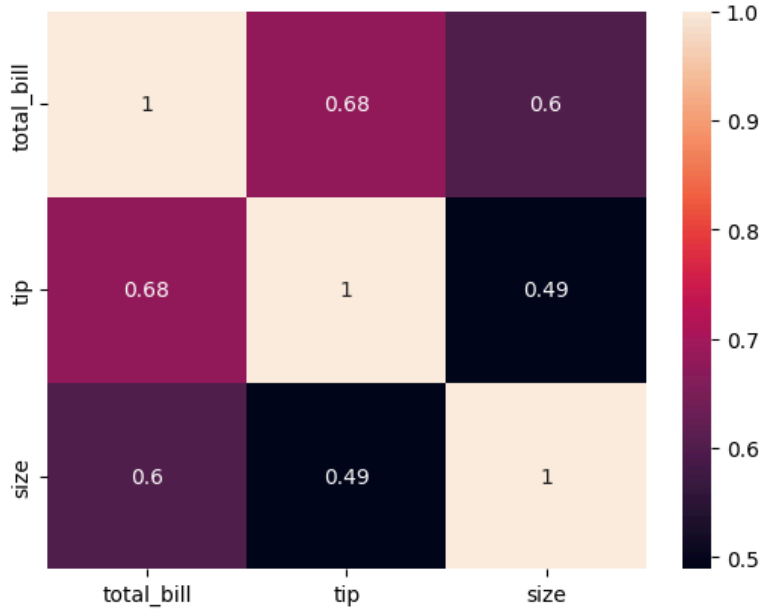

total_bill
tip
10
20
30
40
50
2
4
6
8
10

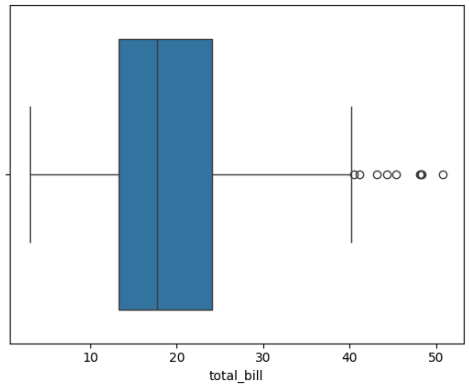


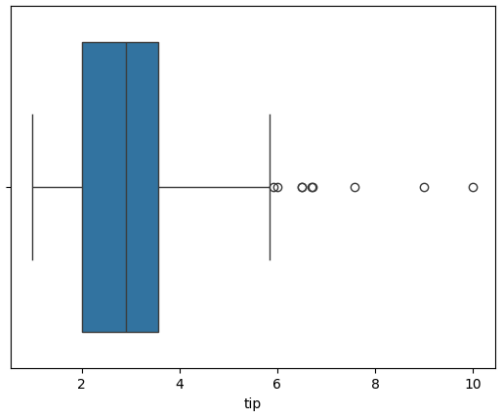

```
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```

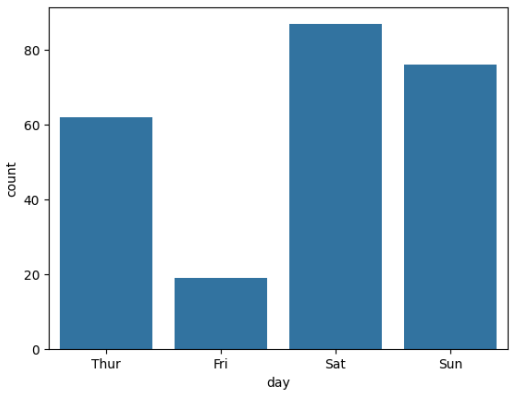


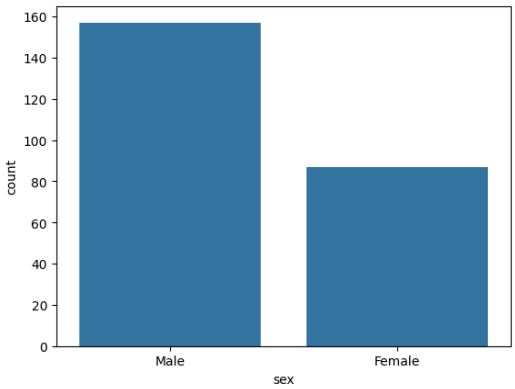

total_bill
tip
size
day
Thur
Fri
Sat
Sun

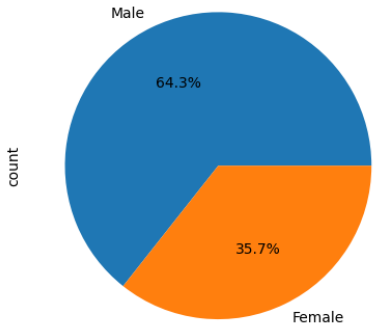

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
1.0
0.9
0.8
0.7
0.6
0.5

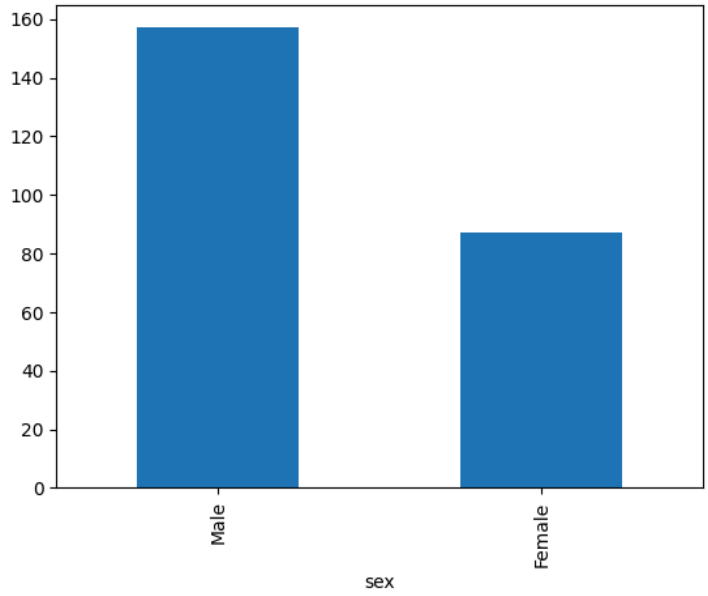

10
20
30
40
50
total_bill

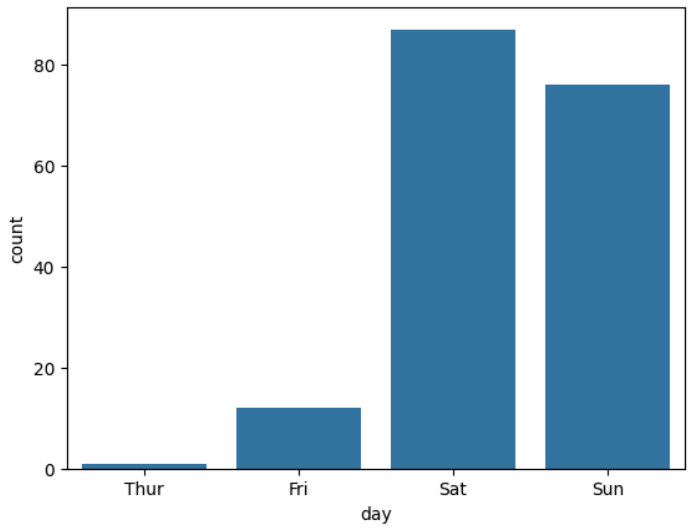

2
4
6
8
10
tip


0
20
40
60
80
count
Thur
Fri
Sat
Sun
day


0
20
40
60
80
100
120
140
160
count
Male
Female
sex


Male
64.3%
35.7%
Female
count


160
140
120
100
80
60
40
20
0
Male
Female
sex


80
60
40
20
0
count
Thur
Fri
Sat
Sun
day

RESULT:

Exploratory Data Analysis (EDA) was successfully performed using quantitative and qualitative analysis. Quantitative analysis helped summarize numerical data, while qualitative analysis helped understand categorical data and identify patterns and insights.